

Empower My Digital Neighbors: How LLM-Driven NPCs Shape Player Interaction in Single-Player and Multiplayer Contexts

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Abstract

Recent advances in large language models (LLMs) have enabled non-player characters (NPCs) in digital games to engage in open-ended natural language interaction, raising expectations for richer and more emotionally engaging player experiences. However, how such LLM-driven NPCs shape player experience in socially situated gameplay remains underexplored. We present an exploratory study of LLM-driven NPCs in *Stardew Valley*, examining how players engage with LLM-driven characters across single-player and multiplayer contexts. Our findings reveal that the impact of LLM-driven NPCs is highly context-dependent: while such NPCs can enrich interaction in single-player play, engagement with NPCs consistently decreased in multiplayer settings, where player-player interaction became dominant. These results suggest that the value of LLM-driven NPCs lies not in competing with social interaction, but in designs that complement existing gameplay dynamics and social relationships.

CCS Concepts

• Human-centered computing → Empirical studies in HCI; • Applied computing → Computer games.

Keywords

Generative AI, Digital Game, NPC, Interactive Design

ACM Reference Format:

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1 Introduction

Large language models (LLMs) have recently attracted growing attention in the design of intelligent systems, including their application in digital games. Within game development, LLMs are increasingly used to generate dialogue [13], personalized storyline [8, 9], NPCs [16], and support emergent narratives [10]. Compared to traditional scripted dialogue systems, LLM-driven NPCs promise greater flexibility, responsiveness, and perceived agency, raising expectations that such characters may foster deeper emotional engagement and richer player experiences [7, 17].

Despite this enthusiasm, existing research has largely focused on the technical capabilities or design possibilities of LLM-driven NPCs, with relatively limited empirical investigation into how players actually interact with such characters during gameplay. In particular, it remains unclear whether LLM-driven NPCs consistently enhance player experience, or whether their effects depend on contextual factors such as players' prior experience or the presence of other human players. While many games are inherently social, few studies have examined how LLM-driven NPCs are perceived when human co-players are present and competing for players' attention and interaction.

To address these gaps, we conducted an exploratory study examining how LLM-driven NPCs influence player experience and interaction willingness in both single-player and multiplayer contexts. We selected *Stardew Valley* as our research platform due to its sandbox-style gameplay, non-mandatory NPC interactions, and strong emphasis on player autonomy. By introducing an LLM-based dialogue mod into the game, we were able to compare player experiences across different conditions while keeping the core game mechanics intact.

This work aims to answer the primary research question that how LLM-driven NPCs influence player experience, particularly player-NPC interaction across different gameplay contexts. By addressing the question, we seek to contribute situated insights into the design of LLM-driven NPCs as interaction partners in socially rich player environments.

2 Related Work

Large language models (LLMs) have become one of the most dominant approaches for driving artificial intelligence systems. In the game domain, LLMs are increasingly adopted as foundational tools for text generation and 2D content creation [11, 12], and are gradually being integrated into game mechanics and interaction systems [6].

A key area of interest is the embedding of LLMs into non-player characters (NPCs). Compared to traditional scripted NPCs, LLM-driven NPCs enable more flexible and open-ended player interaction. Existing work on LLM-driven NPCs can be broadly categorized into three interaction paradigms. First, some studies focus on single LLM-driven NPC–player interaction, where the model is prompted with a predefined personality and backstory, enabling flexible, in-character responses that can provide real-time feedback, guidance, or explanations during gameplay [2, 14]. Second, other researches explore multi-NPC systems, in which multiple LLM-driven agents with distinct identities interact not only with players but also with one another, allowing emergent dialogue and narrative events to arise within the game world [1, 10, 15]. Third, some systems emphasize player-authored NPCs, enabling players to create or customize LLM-driven characters based on personal preferences, thereby granting players direct control over the social composition of the game environment [16].

As a core component of a game’s internal order, NPC–player interaction influences playability across multiple dimensions [3, 4] and has long been a central concern in game design and user experience research. However, in games that incorporate LLMs, existing studies tend to focus either on narrative outcomes or on technical analyses of AI systems. Given the diversity of ways in which LLMs can be embedded into games, relatively little research has directly examined how interaction between players and LLM-driven NPCs affects game playability. This study addresses this gap by focusing on player experience and interaction willingness toward LLM-driven NPCs.

3 Methodology

3.1 Game Context

Stardew Valley is a farming simulation and sandbox game that allows players to freely engage in activities such as farming, exploration, crafting, and social interaction. NPC interaction, while available throughout the game, is largely optional and not strictly required for progression, making it a suitable context for examining voluntary player–NPC engagement. To enable open-ended NPC interaction, we employed *Social Valley*, a third-party mod that integrates an LLM-driven dialogue system into the game. The mod allows players to initiate free-form conversations with NPCs by typing while preserving the original game structure.

3.2 Participants and Experimental Procedure

Thirteen participants were recruited for the study. All participants were categorized as either new players (no or minimal prior exposure to the game) or experienced players based on self-reported gameplay experience. None of the participants had previously used any *Stardew Valley* mods driven by LLMs.

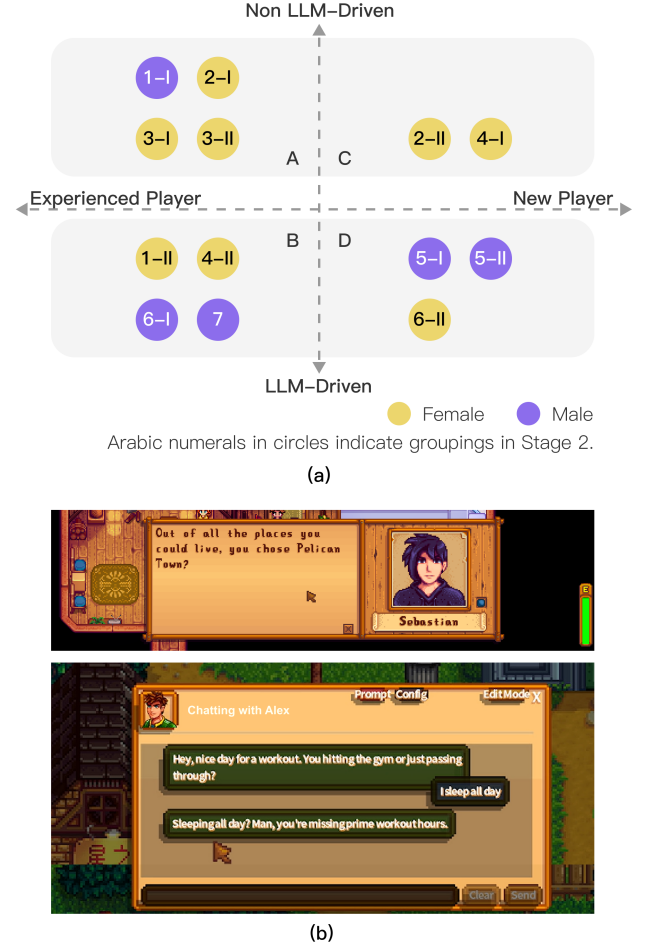


Figure 1: (a) Participant grouping in Stage 1. Participants were divided into four groups based on prior gameplay experience and whether the *Social Valley* mod was installed during the session. The group sizes were as followed, forming a 2×2 factorial design. Group A: 4 experienced players without using the mod. Group B: 4 experienced players using the mod. Group C: 2 new players without using the mod. Group D: 3 new players using the mod. (b) Representative NPC dialogue interactions without (top) and with (bottom) the mod.

The study was conducted in two stages over a period of 5–7 days.

In **Stage 1 (single-player)**, participants played *Stardew Valley* individually under different experimental conditions defined by two factors: (1) prior experience with the game (new vs. experienced), and (2) whether NPCs were powered by the LLM-driven dialogue mod.

In **Stage 2 (multiplayer)**, participants engaged in cooperative gameplay in pairs. Participants were randomly paired, forming a total of seven multiplayer groups. In six groups, both players were recruited participants. In the remaining group, one participant was paired with a non-participating player who had prior experience with both *Stardew Valley* and the LLM-driven NPC mod. To ensure

consistency in data analysis, only data from the recruited participant in this pairing were included in subsequent analyses. Among the seven groups, only one group consisted of two participants who were acquainted prior to the study; all other participants reported no prior contact with their co-players before the multiplayer session. During the session, the two participants in the same group were allowed to talk to each other through an online meeting.

After each stage, participants completed a questionnaire adapted from the Game Experience Questionnaire (GEQ) [5], with questions about players' gameplay experience and social presence. Semi-structured interviews were conducted after the questionnaire to capture participants' subjective interpretations and reflections.

4 Results

4.1 Experience-Dependent Effects of LLM-Driven NPCs

In single-player gameplay, AI integration did not induce fully consistent changes across new and experienced players. However, notable effects emerged on several key dimensions, sometimes with opposite trends between the groups.

In Behavioral Involvement, experienced players' average scores decreased after AI integration, while new players' scores increased. In Tension, the pattern was reversed. For Empathy, new players showed a clear increase in scores, whereas experienced players' mean scores declined, though medians remained stable. The large range between minimum and maximum values indicates substantial individual variation among experienced players. Interview data suggested that these players often held established expectations regarding NPC behavior and narrative roles; deviations introduced

by the LLM-driven dialogue sometimes disrupted their sense of coherence within the game world.

By contrast, in Negative Feelings, Immersion, and Competence, mean differences across conditions were small, suggesting these dimensions were less sensitive to the experimental manipulation and generally stable.

These findings suggest that the perceived value of LLM-driven NPCs is moderated by prior player experience rather than being uniformly positive.

4.2 Human Co-Players Marginalize NPC Interaction

In multiplayer gameplay, all other dimensions either decreased or remained stable, except for Negative Affect. Over 60% of participants reported reduced NPC interaction, and two did not interact with NPCs at all in Stage 2. Nearly half adjusted their gameplay due to the presence of human co-players.

Overall, the rise in Negative Affect and the decline in Negative Feelings and Tension likely reflect weakened player–NPC interactions, whereas decreases in other positive dimensions may reflect high acceptance of player–player interactions, with interview data suggesting that participants tended to reckon human co-player being more active and psychologically closer.

This pattern was consistent regardless of players' prior experience with the game, suggesting that the presence of human co-players fundamentally reshapes social attention within the game. Even though NPCs were powered by an LLM and capable of open-ended conversation, they were rarely treated as primary interaction partners in multiplayer contexts.

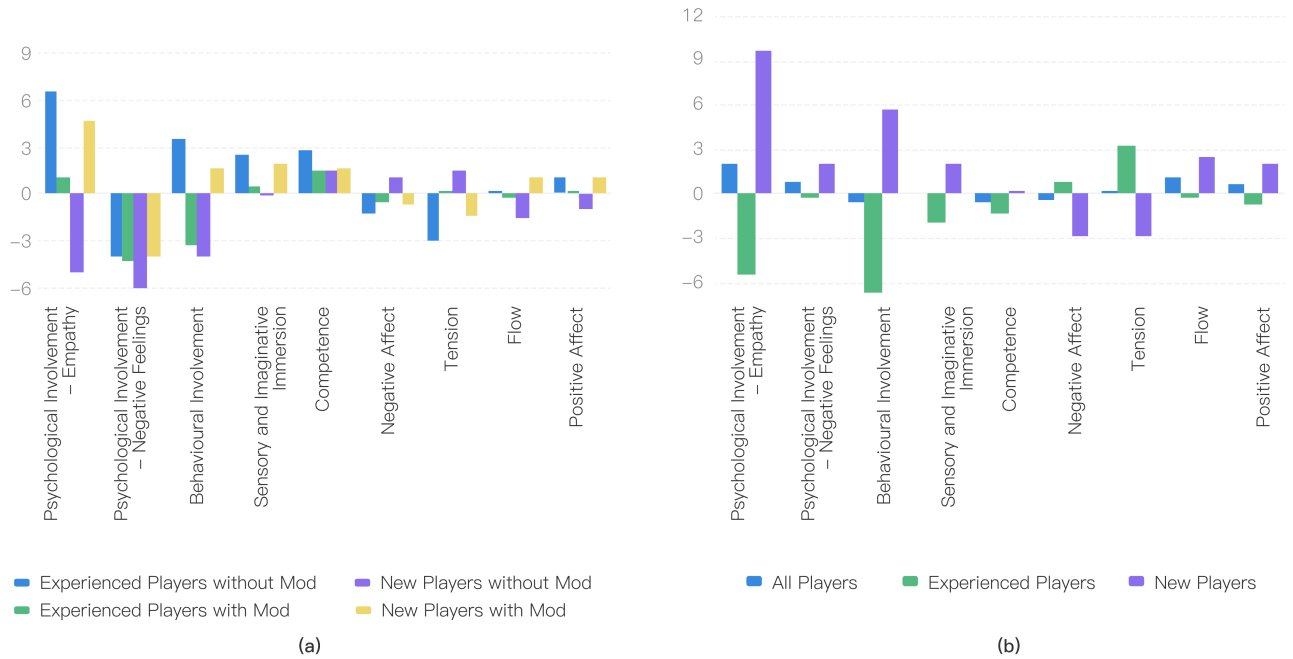


Figure 2: (a) The average scores on the 9 dimensions of GEQ for different participant conditions (prior gameplay × AI integration) in the first experimental round. (b) The difference value between AI and non-AI conditions across different participant groups.

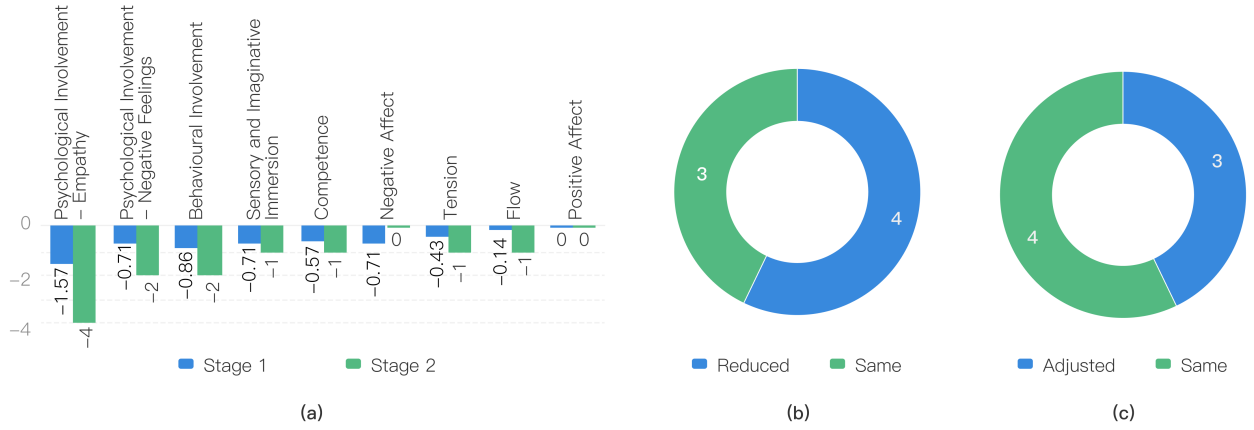


Figure 3: Comparison of game states for the seven participants who used the mod in both stages. (a) The differences in NPC-related questionnaire scores between Stage 1 and Stage 2. (b) and (c) depict, respectively, participants’ semi-structured interview responses on changes in NPC interaction frequency in multiplayer mode and on adjusting in-game activities (frequency of player-NPC interaction, farming, mining, etc.) due to the presence of leveling partners.

5 Discussion

Our findings highlight the situated and context-dependent nature of the value of LLM-driven NPCs. Rather than uniformly enhancing player experience, LLM-driven NPCs interact with players’ expectations, prior experience, and social context in complex ways. For new players, AI-enhanced NPCs may lower barriers to engagement and foster early emotional connections, leading to increased behavioral involvement and reduced tension. In contrast, for experienced players, the same features may conflict with established mental models of the game world, resulting in more ambivalent or even negative experiences.

Results from multiplayer gameplay further demonstrate the limits of social attention. When human co-players are present, they tend to dominate interaction bandwidth, marginalizing NPCs regardless of their AI capabilities. This suggests that designing NPCs as direct substitutes for human social partners is unlikely to be effective in socially rich gameplay contexts.

Overall, LLM-driven NPCs appear most beneficial in single-player or low-social environments, where they can function as companions, guides, or sources of emotional response. In multiplayer settings, NPCs are better positioned as complementary agents that support human players through guidance, information integration, or pacing and task coordination. Notably, case observations indicate that NPCs may be perceived as “intelligent assistants” when verbal communication between players is limited, highlighting their potential value in supportive, non-competitive roles.

From a design perspective, these findings underscore the importance of complementarity rather than replacement. Instead of competing with human players for attention or emotional engagement, LLM-driven NPCs should be designed to augment collaboration, provide contextual assistance, and enrich the game world in ways that align with existing social dynamics.

6 Limitations and Future Work

This study is exploratory in nature and subject to several limitations. First, the sample size was relatively small and consisted primarily of young adult participants (Aged 18-30), limiting the generalizability of the findings. Second, the relatively short study duration captured primarily short-term reactions rather than long-term adaptation. Thirdly, the study focused on one single game with multiple play-style, so players’ motivation to engage with NPCs may be weakened by other factors. Consequently, conclusions regarding player-NPC interaction in this study are partly shaped by the game’s genre and design structure, and their applicability to narrative- or character-driven games remains to be further examined.

Additionally, though player-NPC and player-player interactions were mediated through different modalities, this setup reflects common interaction practices in *Stardew Valley* and similar games, where NPC interaction is typically interface-based while co-players communicate verbally. As voice-based NPC interaction becomes more feasible, future work should further examine how interaction modality shape player attention and engagement across social contexts.

Overall, future research should systematically examine how LLM-driven NPCs can be functionally differentiated across gameplay contexts, player backgrounds, social configurations, and interaction modality, in order to better integrate LLM technologies into game ecosystems in ways that meaningfully enhance playability and player experience.

7 Conclusion

We present an exploratory mixed-methods study examining how LLM-driven NPCs shape player experience across single-player and multiplayer contexts in *Stardew Valley*. Our findings show that the effects of LLM-driven NPCs are not uniform but depend on player experience and social configuration. AI-enhanced NPCs increased empathy and interaction willingness among new players in single-player gameplay, while producing mixed responses from

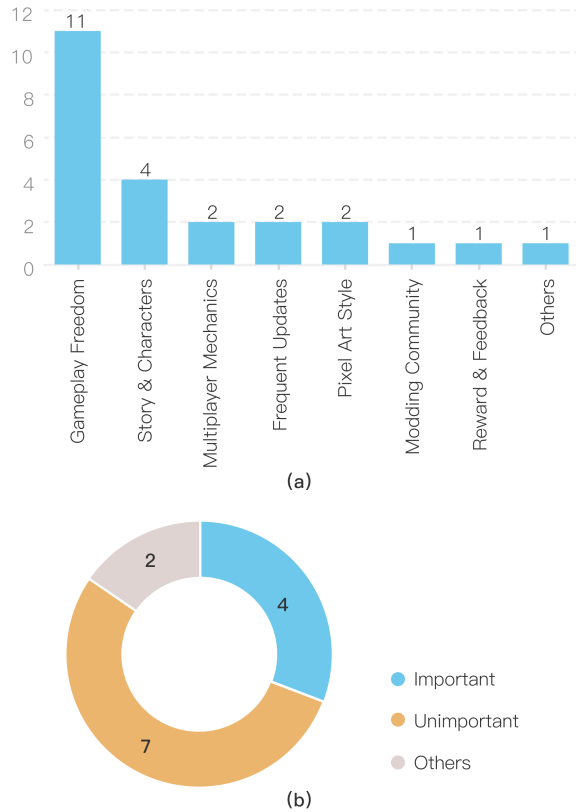


Figure 4: Semi-structured interviews indicated that most participants (11/13) identified gameplay freedom as the primary source of the game's appeal, while only a small number (4/13) emphasized NPC-related narratives or dialogues. Moreover, over half of the participants reported that interactions with NPCs occurred mainly through incidental encounters and were perceived as supplementary rather than central to core gameplay activities such as management and exploration.

experienced players. In multiplayer settings, player–NPC interaction consistently decreased as human–human interaction became dominant.

This work contributes empirical evidence that the value of LLM-driven NPCs is context-sensitive rather than universally beneficial. We highlight design implications for positioning intelligent NPCs as complementary agents whose roles adapt to gameplay structure and social context.

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